

AIR POLLUTION CONTROL - FIELD INSPECTION WORK SHEET

Line	County			Source			Emission			Action			Date			Agency		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Line 1	01	10	01	02	22	02	00	00	00	00	00	00	00	00	00	00	00	00
	1	3	4	7	8	12	13	15	16	18	52	54	59	66	63	33		

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

00151 31 101

16 18 78 80

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Company: Martin-Marietta Cement, Western Division SIC: 3241
 Address: PO Box 529 Lyons, Colorado 80540 County: Boulder
 Person Contacted: Gerald Anderson Phone: (303) 534-4206
 Inspected by: George Matthews Time: 1:30 - 3:30 - PM/PM

1. Annual inspection per SHD Contract.
Clear day; ambient air temp. ~50°F; winds westerly, est. < 5 mph.
2. Second site visit made 3-12-81, 2:00 - 3:15 pm, to obtain more information about 1981 production levels and fuel consumption. On this visit, the entire production plant was closed for maintenance, including relining the kiln w/ refractory brick and baghouse cleaning. An interior tour of the length of the kiln, of the clinker cooler, and the kiln baghouse was made.

3. Point 001: Kiln Stack.

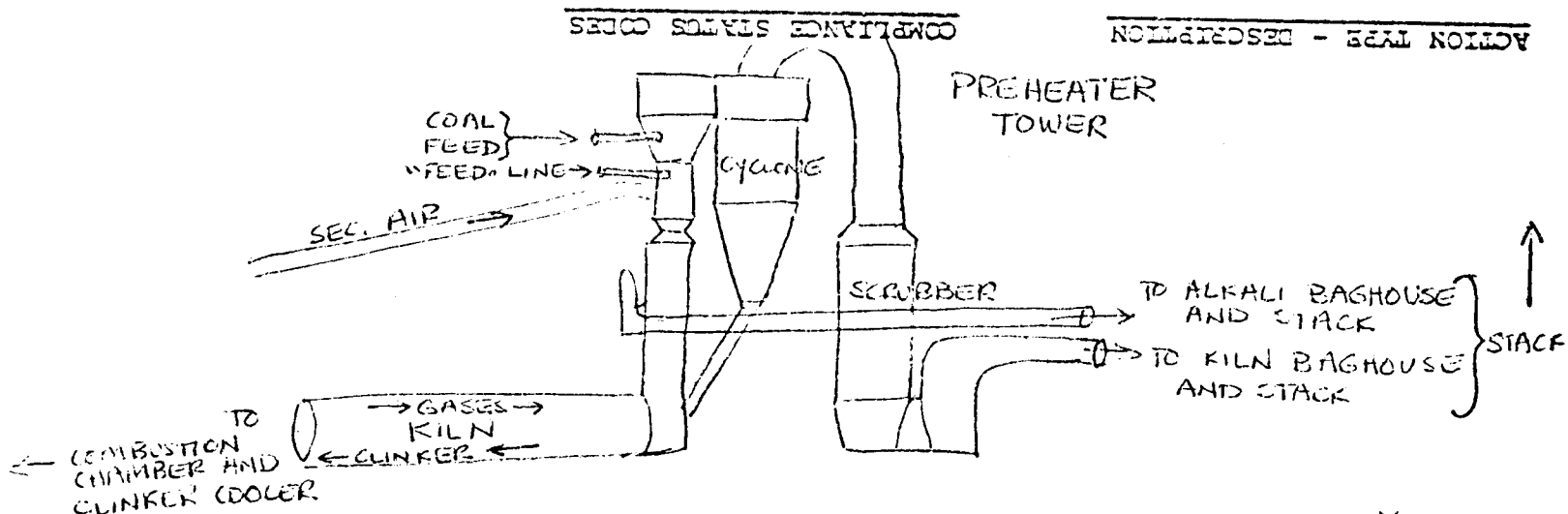
System components of this emission point include:

- a. Combustion chamber: 60% of the plant's pulverized coal and the "feed" from the kiln feed or homogenizing silos go into this chamber or pre-calciner. The feed is a formulation of limestone or shale, iron ore, and sand.
- b. Cyclone: This deposits the pre-heated, calcined material into the kiln.
- c. Wet Scrubber Spray Tower: City of Longmont pretreated water is used at a rate of 150 gpm. The water cools the exhaust gases, locking SO₂ into the dust. All injected water evaporates.
- d. Kiln Baghouse: Waste dust is taken to waste.
- e. Alkali Baghouse: piles in quarry reclamation process.
- f. Stack: Formal visibility emission observation not conducted.

At 19 (Rev. 7-77-10) 0% Opacity observed

- 12 - Inspection
22 - Source Test
23 - Opacity Observation
24 - Odor Observation

- 0 - Unknown
1 - Not in Compliance
2 - In Compliance by Source Test
3 - In Compliance by Inspection
4 - In Compliance by Certification
5 - In Compliance with Increments of Progress
6 - Not in Compliance with Increments of Progress
7 - Unknown Compliance with Increments of Progress
8 - No Applicable State Regulation
9 - In Compliance -- Closed Down



3.f. Slack continued! In-stack meters are installed for opacity, SO_2 and HC.

— 1981 Production : 424,000 T cement

Fuel Consumption of Kiln: 322,278 MCF natural gas
2,030,000 gal #6 fuel oil
64,548 T coal

Energy conversions = 4,646,919 BTU/ton of clinker

The amounts of time for each type of fuel usage are not individually known.

Calculations

Point	Pollutant	Regulations	Allowable Em	Actual Emissions
001: Kiln Stack	TSP	Reg. No. 11A. Permit No C-12,444-2. (102 T/yr) HP42, 8.6	102 T/yr	$(424,000 \text{ T cement/yr}) (245 \text{ lbs TSP/T cement}) (0.001)$ $(2000 \text{ lbs TSP/T TSP})$ = 52 T of TSP in 1981

Compliance Status: By calculation and by observation. In compliance.

Martin-Marietta Cement, Western Division
Inspection Report for 2/16/82

3: Point 003 (cont.)

Calculations

<u>Pollutant</u>	<u>Regulation</u>	<u>Allowable Em.</u>	<u>Actual Emissions</u>
SO ₂	Reg. No. 1, III A 4 a ii Permit C-12,444-2 AP42, 2.6-4	1.2 lb/10 ⁶ Btu and 851 T/yr	Product: $\frac{(424,000 T_{\text{Cement}})(10.2 \text{ lb SO}_2 / T_{\text{Cement}})}{2000 \text{ lb SO}_2 / T_{\text{SO}_2}} = 2162 T$

Fuel Combustion

The percentages of year devoted to @ type of fuel use were not determined. If @ type of fuel had been used year round, following em. calc's apply.

Oil

$$\frac{(424,000 T_{\text{Cement}})(4.2 \text{ lb SO}_2 / T_{\text{Cement}})(1\% \text{ Sulfur})}{2000 \text{ lb SO}_2 / T_{\text{SO}_2}} = 890 T$$

Coal

$$\frac{(424,000 T_{\text{Cement}})(6.8 \text{ lb SO}_2 / T_{\text{Cement}})(0.5\% \text{ Sulfur})}{2000 \text{ lb SO}_2 / T_{\text{SO}_2}} = 721 T$$

Because SO₂ emissions w/ oil are higher, assume oil use year round for sake of calculations

Product SO₂ : 2162 T
 Fuel SO₂ : 890 T
 3052 T

Per AP42, 2.6-4, 75% SO₂ reduction can be expected

$$(3052 T)(0.25) = 756 T$$

 of SO₂ in 1981

Status: In compliance by calculation

Martin-Marietta Cement
Insp. Report for 2/16/82.

3. Point 001 (cont)

Pollutant	Regulation	Allowable Emission	Actual Emission
CO	None	None specified	10 T. from 8-21-80 insp. report
HC	None	None specified	5 T. from 8-21-80 insp. report
NO _x	None	None specified	$\frac{(424,000 \text{ T}_{\text{cement}})(2.6 \text{ lb NO}_x/\text{T}_{\text{cement}})}{2000 \text{ lb NO}_x/\text{T}_{\text{on NO}_x}} = 551 \text{ T}$

AP42, 8.6-3

4. Point 002 : Raw material crushers : 600 T/hr capacity. Baghouse control.
Belts have capacity of 150 T/hr; all belts are equipped with canopies.
No fugitive dust noted.

Point	Pollutant	Regulation	Allowable Em	Actual Emissions
002	TSP	Reg. No. I A1 20% opacity Reg. No. II C1 Fig 2: AP42, 8.20	50 lb/hr	$(600 \text{ T}_{\text{rock}}/\text{hr})(0.5 \text{ lb TSP}/\text{T}_{\text{rock}})(0.01) = 3 \text{ lb/hr}$

Status: In compliance by observation and by calculation.

5. Point 003 : Raw material drier: Crushed limestone is dried to reduce water content to <10%. Fuel is natural gas, $149,211 \times 10^6 \text{ ft}^3$ used in 1981.
Baghouse control.

Point	Pollutant	Regulation	Allowable Em.	Actual Emission
003	TSP	Reg. No. I A1 Reg. No. II C Permit C-12,444-1 AP42, 8.6-3 AP42, 1.4-2	20% 64 lb/hr = 144 T/yr	$\frac{(424,000 \text{ T}_{\text{cement}})(96 \text{ lb TSP}/\text{T}_{\text{cement}})(0.001)}{(2000 \text{ lb TSP}/\text{T}_{\text{TSP}})} = 20.4 \text{ T}$

Fuel

$$\frac{(149,211 \times 10^6 \text{ ft}^3)(10 \text{ lb TSP}/10^6 \text{ ft}^3)(0.001)}{(2000 \text{ lb TSP}/\text{T}_{\text{TSP}})} = 0.7 \text{ T}$$

Total TSP in 1981 : ~21 T

Status: In compliance by calc.

Martin Marietta, Western Division
Inspection Report for 2/16/82

5. Point 003 (cont)

Pollutant	Regulation	Allowable Emission	Actual Emission
SO ₂	Reg. No 1, III S or Permit C-12,144-1 (when using oil) AP 42, 1.4-2	2T/day	$(149,211 \times 10^6 \text{ ft}^3/\text{yr})(1\text{ yr}/365 \text{ days}) = 409 \times 10^6 \text{ ft}^3/\text{day}$ $(409 \times 10^6 \text{ ft}^3/\text{day})(0.6 \text{ lb SO}_2/10^6 \text{ ft}^3) = 0.12 \text{ T/day}$ $\frac{0.12 \text{ T/day}}{2000 \text{ lb SO}_2/\text{T SO}_2} = 44 \text{ T/yr}$

Pollutant	Regulation	Allowable Emission	Actual Emission	Status
CO	None AP 42, 1.4-2	None Specified	$(149,211 \times 10^6 \text{ ft}^3/\text{yr})(131 \text{ lb HC}/10^6 \text{ ft}^3) = 1268 \text{ T/yr}$ $\frac{1268 \text{ T/yr}}{2000 \text{ lb HC}/\text{T HC}}$	In compliance
HC	"	None specified	$(149,211 \times 10^6 \text{ ft}^3/\text{yr})(31 \text{ lb HC}/10^6 \text{ ft}^3) = 224 \text{ T/yr}$ $\frac{224 \text{ T/yr}}{2000 \text{ lb HC}/\text{T HC}}$	
NO _x	"	None specified	$(149,211 \times 10^6 \text{ ft}^3/\text{yr})(120 \text{ lb NO}_x/10^6 \text{ ft}^3) = 13428 \text{ T/yr}$ $\frac{13428 \text{ T/yr}}{2000 \text{ lb NO}_x/\text{T NO}_x}$	

NOTE: The baghouse control for this drier was moved from the clinker cooler. Martin-Marietta was informed that final approval of Permit No. C-12,444 was being withheld pending a performance test of the dryer w/ baghouse, Part 1 of the permit.

Point 003 should be put on EIS.

6. Point 004: A-frame storage. Reg No. 1. II'D

Small baghouses at transfer points inside the building. Only fugitive dust points would be through doors. No fugitive dust problems noted. Status: In compliance.

7. Point 005: Site Fugitive Dust. Reg No. 1. II'D.

While production crews work 7 days/week, site clean-up crews work only 1 shift/day, 5 days/week. So, the potential for fugitive dust problems exists primarily on Sat, Sun, Mon given sufficient wind velocities. Initial inspection was performed after a long weekend and there was considerable dust buildup on runways, etc., but because there was no wind, no fug. dust problem was noted. Status: In compliance at time of inspection.

8. Point 006: Coal Handling Reg. No. 1, IFD.

Bituminous low-sulfur coal (0.5% S) from Colo. Wild, Craig Colo was used in 1981. Coal is pulverized in mill within silo. Only dust problems would be through doorways. No problems noted at time of inspection. The coal baghouse was removed Feb. or March 1981 due to recurring problems. Now all coal dust is recirculated into system.

Status: In compliance.

9. Point 007: Quarries. Reg. No. 1, IFD. Two active limestone pits and one sand quarry, which was not toured) are on site. Haul roads are watered and cml. dust stabilizer, lignin sulfate, is also applied. Haul trucks generally travelling ≤ 30 mph. Waste materials are deposited into quarry excavations. Because quantity is about 50% greater than excavated material removed, elevations increase appreciably. Some evidence that revegetation has not taken in all places. Soil crust well formed. No fugitive dust noted at original inspection on 2/16/82. Wind on 3/12/82 was gusting to 50 mph + and there were dust problems. Water and stabilizers were being vigorously applied.

Status: In compliance.